

Analog Lab Report Week 4

Ananya Tiwari

Roll no. – 24f3100260

Objective :

This experiment aims to design, derive, and implement two band pass filter of two different centre frequencies. We have to derive the equation for circuit and find the value for the resistors and capacitors.

List of Components :

1. 1 – MCP6004
2. Capacitors and resistors
3. Power Supply (ADALM 1000)
4. Breadboard

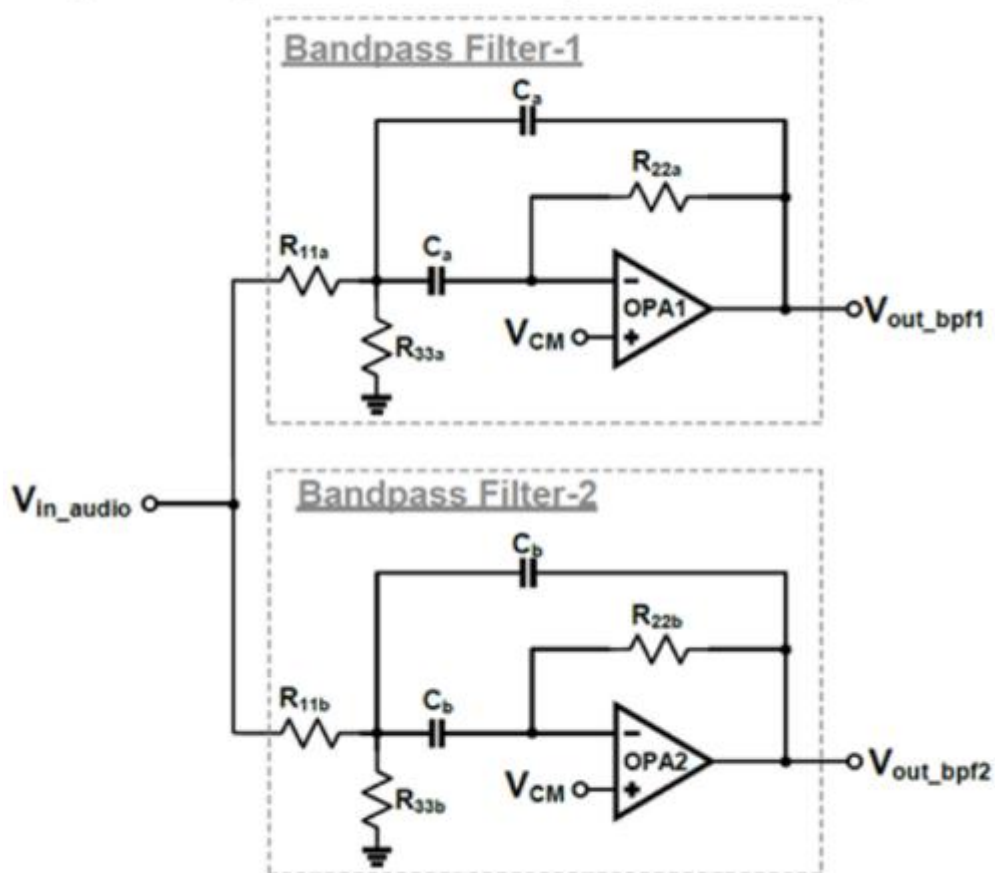
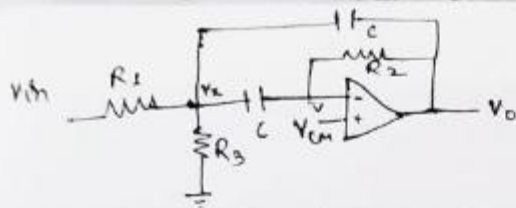


Figure 8 Schematic diagram of a bandpass filter.

Derivation of Formula :



2413100260

Aanya Tiwari

$$V_o = -\frac{R_2}{R_3} V_x$$

$$\frac{V_x - V_i}{R_1} + \frac{V_x}{R_3} + \frac{V_x - V_o}{R_2} + \frac{V_x}{R_2} = 0$$

$$\frac{V_x (R_2 + R_1)}{R_1 R_3} - \frac{R_2 V_i}{R_1 R_3} + \frac{2V_x - V_o}{R_2} = 0$$

$$V_x (R_2 + R_1) R_2 - R_2 R_3 V_i + 2R_1 R_3 V_x - R_1 R_3 V_o = 0$$

$$V_x [(R_2 + R_1) R_2 + 2R_1 R_3] = + R_2 R_3 V_i + R_1 R_3 V_o$$

$$V_x = \frac{R_2 R_3 V_i + R_1 R_3 V_o}{(R_2 + R_1) R_2 + 2R_1 R_3}$$

$$V_o = -\frac{R_2}{R_3} \times \frac{R_2 R_3 V_i}{(R_2 + R_1) R_2 + 2R_1 R_3} + \frac{R_2}{R_3} \times \frac{R_1 R_3 V_o}{(R_2 + R_1) R_2 + 2R_1 R_3}$$

$$V_o = -\frac{R_2 R_3 V_i}{(R_2 + R_1) R_2 + 2R_1 R_3} - \frac{R_1 R_2 R_3 V_o}{R_3 [(R_2 + R_1) R_2 + 2R_1 R_3]}$$

$$V_o \left[1 + \frac{R_1 R_2 R_3}{R_3 [(R_2 + R_1) R_2 + 2R_1 R_3]} \right] = -\frac{R_2 R_3 V_i}{(R_2 + R_1) R_2 + 2R_1 R_3}$$

$$V_o \left[\frac{R_3 [(R_2 + R_1) R_2 + 2R_1 R_3] + R_1 R_2 R_3}{R_3} \right] = -R_2 R_3 V_i$$

$$V_o = -\frac{R_2 R_3 R_3 V_i}{(R_3 [(R_2 + R_1) R_2 + 2R_1 R_3] + R_1 R_2 R_3)}$$

$$V_o = \frac{-R_2 R_3 X_c V_i}{(R_3 + R_1) X_c^2 + 2 R_1 R_3 X_c + R_1 R_2 R_3}$$

$$V_o = \frac{-R_2 R_3 (1/X_c) V_i}{((R_3 + R_1)(1/X_c)^2 + 2 R_1 R_3 (1/X_c) + R_1 R_2 R_3)}$$

$$= \frac{-R_2 R_3 (1/X_c) V_i}{(1/X_c)^2 ((R_3 + R_1) + 2 R_1 R_3 X_c + R_1 R_2 R_3 (X_c)^2)}$$

$$= \frac{-R_2 R_3 X_c V_i}{[X_c^2 R_1 R_2 R_3 + 2 R_1 R_3 X_c + (R_3 + R_1)]}$$

$$= \frac{-R_2 R_3 X_c}{X_c^2 R_1 R_2 R_3} \left[X^2 + \frac{2 R_1 R_3 X_c}{X_c^2 R_1 R_2 R_3} + \frac{(R_3 + R_1)}{X_c^2 R_1 R_2 R_3} \right]$$

$$\omega_o^2 = \frac{R_3 + R_1}{X_c^2 R_1 R_2 R_3} \Rightarrow \left\{ \omega_o = \frac{1}{X_c} \sqrt{\frac{(R_3 + R_1)}{R_1 R_2 R_3}} \right\}$$

$$\omega_o = \frac{2 R_1 R_3}{X_c R_1 R_2 R_3} \Rightarrow \frac{1}{X_c} \sqrt{(R_3 + R_1)} = \frac{1}{X_c} = \frac{2 R_1 R_3}{X_c R_1 R_2 R_3}$$

$$\frac{1}{Q} \times \sqrt{(R_3 + R_1)} = \frac{2R_1 R_3}{\sqrt{R_1 R_2 R_3}}$$

$$\left\{ \frac{\sqrt{(R_3 + R_1)(R_1 R_2 R_3)}}{2R_1 R_3} = Q \right\}$$

$$A_0 \omega_0 \Rightarrow A_0 \left(\frac{2R_1 R_3}{\cancel{C R_1 R_2 R_3}} \right) = \frac{R_2 R_3}{\cancel{C R_1 R_2 R_3}}$$

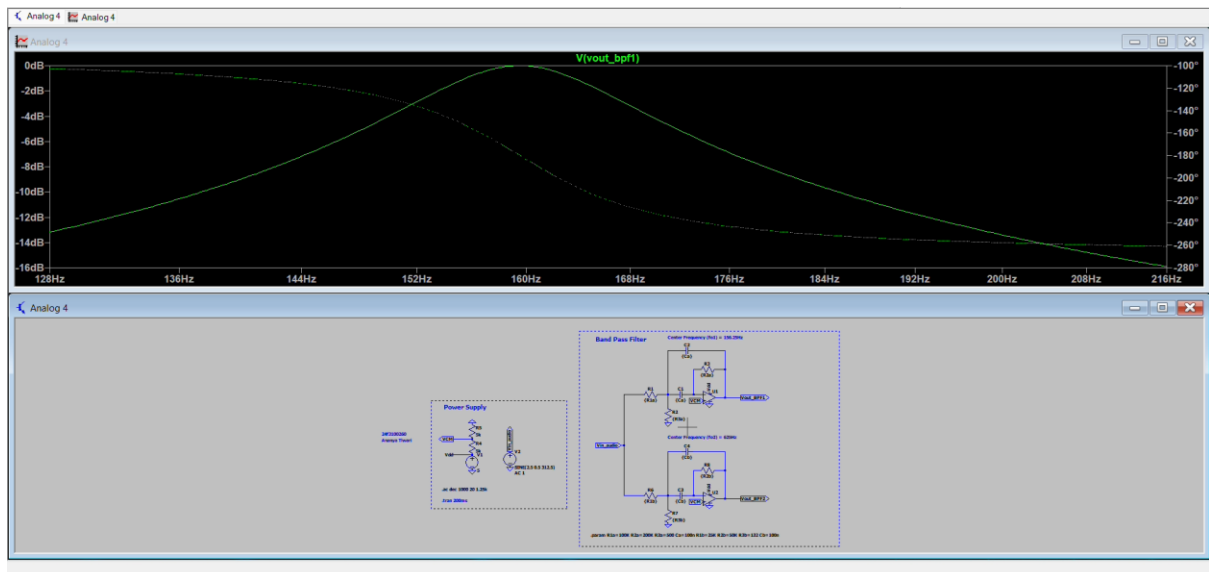
$$\left\{ A_0 = \frac{R_2}{2R_1} \right\}$$

$$\text{for } f_0 \Rightarrow 2\pi f_0 = \frac{1}{C \sqrt{R_1 R_2 R_3}} \sqrt{(R_3 + R_1)}$$

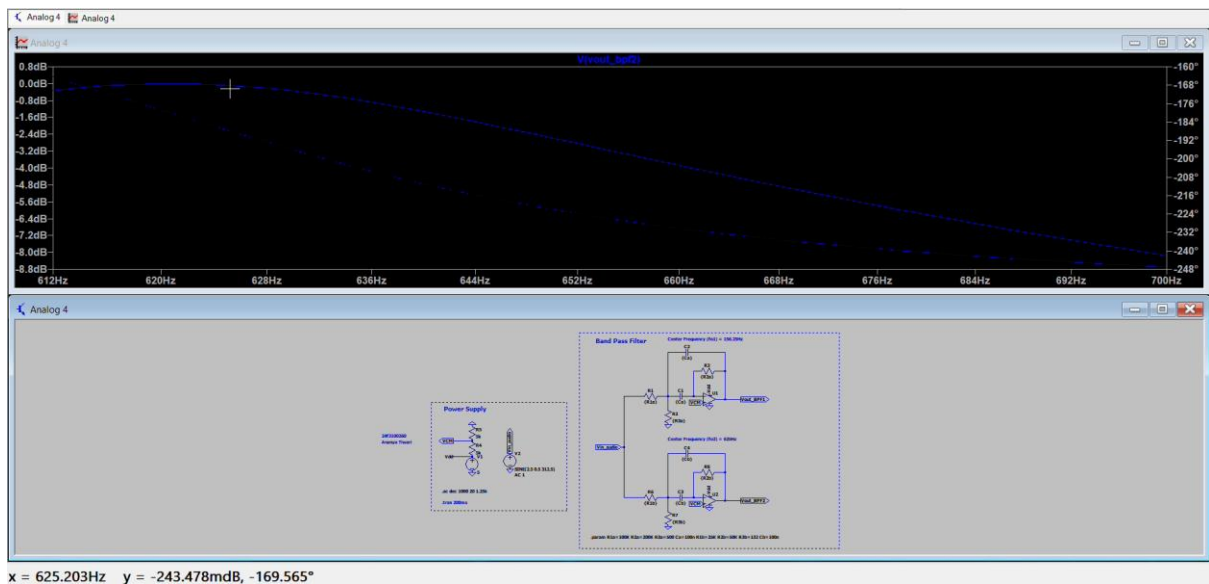
$$f_0 = \frac{1}{2\pi C \sqrt{R_1 R_2 R_3}} \sqrt{(R_3 + R_1)}$$

LTspice Schematics and Output :

Vout of Band Pass Filter 1

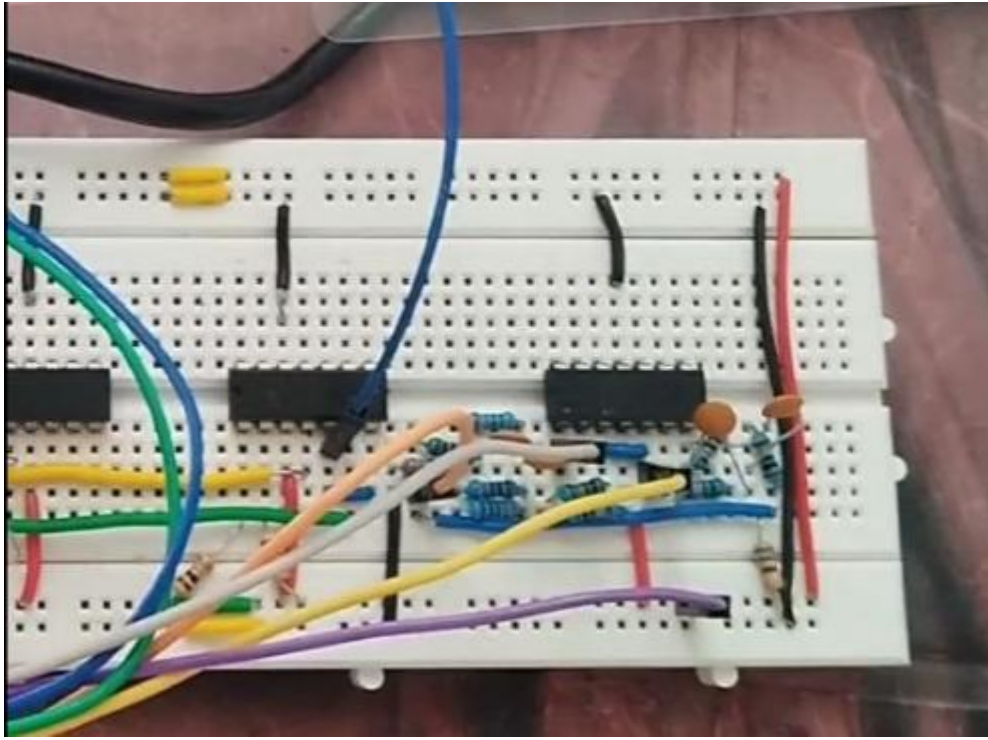


Vout of Band Pass Filter 2



x = 625.203Hz y = -243.478mdB, -169.565°

Physical Circuit :



Physical Circuit Wire Connections :-

Green wire and Blue wire – GND

Yellow – 2.5V

Orange wire – Channel B (*input*)

White wire – Channel A (*output*)

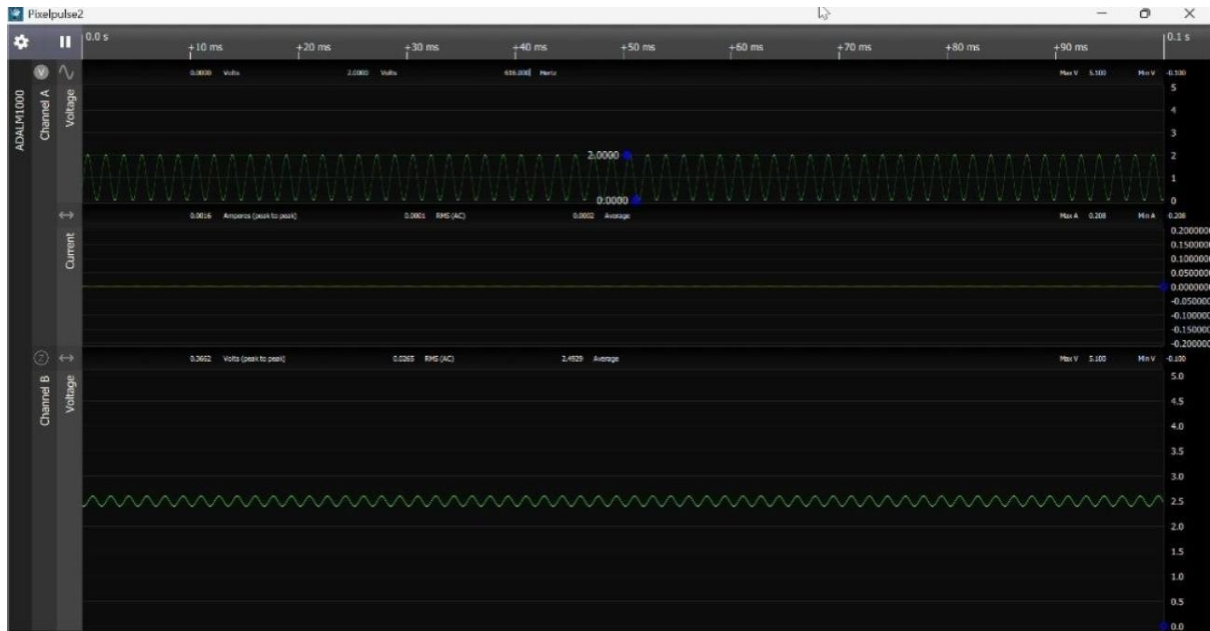
Purple wire – 5V

Physical Circuit Output :-



First one is Vout of Band Pass Filter 1 while the signal frequency is around 156.25 Hz.

Second one is Vout of Band Pass Filter 2 while the signal frequency is around 625 Hz.



Conclusion :-

The experiment was successful, with the physical hardware output validating the simulation models. The output variations are likely due to resistor tolerance factors, although other circuit errors cannot be ruled out.